

SKILL-8

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SEC-8A

SKILL-8:

Aim: Develop a C program to demonstrate environment-variable expansion and handling of selected built-in shell commands.

The program accepts a variable name such as **\$HOME**, obtains its value using **getenv()**, and supports built-in commands **pwd**, **cd**, and **exit** through a dedicated command-execution function.

Procedure :

1. Create the C source file **Skill-8.c**.
2. Read a variable name such as **\$HOME** and remove the leading **\$** before calling **getenv()**.
3. Display the variable value when it is defined.
4. Read a built-in command from the user.
5. Recognize **pwd**, **cd**, and **exit** using **strcmp()**.
6. Execute the corresponding built-in behavior and display the result.

Output :

```
Skill > c Skill-8.c
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4
5  void expand_variable(char *input)
6  {
7      char variable[100];
8
9      if (input[0] == '$')
10     {
11         strcpy(variable, input + 1);
12
13         char *value = getenv(variable);
14
15         if (value != NULL)
16             printf("%s = %s\n", variable, value);
17         else
18             printf("%s = Undefined variable\n", variable);
19     }
20 }
21
22 void execute_builtin(char *command)
23 {
24     if (strcmp(command, "pwd") == 0)
25     {
26         printf("Built-in command: pwd\n");
27         system("pwd");
28     }
29     else if (strcmp(command, "cd") == 0)
30     {
31         printf("Built-in command: cd\n");
32         printf("cd command detected. State maintained.\n");
33     }
34     else if (strcmp(command, "exit") == 0)
35     {
36         printf("Built-in command: exit\n");
37     }
38 }
```

The submitted execution enters **\$HOME**. The program resolves it to **/home/vinay**. It then accepts the built-in command **pwd** and displays the current working directory as **/mnt/c/Testing/OSSP_Testing/skill**.

Code - Main Program:

```
38     else
39     {
40         printf("Invalid built-in command: %s\n", command);
41     }
42 }
43
44 int main()
45 {
46     char input[100];
47
48     printf("=== Variable Expansion ===\n");
49
50     printf("Enter a variable (example: $HOME): ");
51     scanf("%s", input);
52
53     expand_variable(input);
54
55     printf("\n=== Built-in Command ===\n");
56
57     printf("Enter a command (pwd/cd/exit): ");
58     scanf("%s", input);
59
60     execute_builtin(input);
61
62     return 0;
63 }
```

Code - Variable Expansion and Built-in Commands:

```
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing/skill$ ./skill_8
=== Variable Expansion ===
Enter a variable (example: $HOME): $HOME
HOME = /home/vinay

=== Built-in Command ===
Enter a command (pwd/cd/exit): pwd
Built-in command: pwd
/mnt/c/Testing/OSSP_Testing/skill
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing/skill$ |
```

Functions and Program Operations

Function / Operation	Purpose
getenv()	Obtains the value of an environment variable.
strcmp()	Compares the entered command with supported built-in commands.
system()	Executes the pwd command in the demonstrated implementation.
expand_variable()	Processes a variable name and displays its environment value.
execute_builtin()	Handles the supported built-in commands pwd, cd, and exit.

Observation :

- The program performs variable expansion by copying the variable name without the leading **\$** and calling **getenv()**.
- The submitted input **\$HOME** produces **HOME = /home/vinay**.
- The program provides built-in handling for **pwd**, **cd**, and **exit**.
- The submitted **pwd** execution displays the current working directory.
- The demonstrated **cd** branch reports that the command was detected and state is maintained.

Result :

The SKILL-8 program successfully demonstrates environment-variable expansion and built-in command handling. The submitted execution correctly expands **\$HOME** to **/home/vinay** and executes the built-in **pwd** command to display the current directory.

SKILL-8 Completed Successfully